John Fayez
Bash Lab 3

1. Write a script called mycase, using the case utility to check the type of character entered by a user:
    a. Upper Case.
    b. Lower Case.
    c. Number.
    d. Nothing.

```
#!/bin/bash

if [ $# -eq 0 ]; then
        echo "Nothing is passed"

elif [ $# -eq 1 ]; then

case "$1" in
        [a-z])
                echo "$1 is lowercase";;
        [A-Z])
                echo "$1 is uppercase";;
        [0-9])
                echo "$1 is number";;
        *)
                echo "other character is passed";;
esac
fi
```

```
> vi mycase.sh
> chmod +x mycase.sh
> ./mycase.sh
Nothing is passed
> ./mycase.sh a
a is lowercase
> ./mycase.sh A
A is uppercase
> ./mycase.sh 9
9 is number
> ./mycase.sh 9124ASDF
other character is passed
```

2. Enhanced the previous script, by checking the type of string entered by a user:

a. Upper Cases.
b. Lower Cases.
c. Numbers.
d. Mix.
e. Nothing.

```
#!/bin/bash

shopt -s extglob

if [ $# -eq 0 ]; then
        echo "Nothing is passed"
elif [ $# -eq 1 ]; then

case "$1" in
        +([a-z]))
                echo "String is all lowercases";;
        +([A-Z]))
                echo "String is all upercases";;
        +([0-9]))
                echo "String is all numbers";;
        +([a-zA-Z0-9]))
                echo "String is a mix";;
        *)
                echo "Sting is not valid";;
esac

fi
~
"mycase-enhanced.sh" 22L, 341B
```

```
> ./mycase-enhanced.sh
Nothing is passed
> ./mycase-enhanced.sh AAAAAAAAa
String is a mix
> ./mycase-enhanced.sh AAAAAAAA
String is all upercases
> ./mycase-enhanced.sh aaaaaaaabavaskjdf
String is all lowercases
> ./mycase-enhanced.sh 298357
String is all numbers
> ./mycase-enhanced.sh 3
String is all numbers
> ./mycase-enhanced.sh ASD4
String is a mix
>
> ./mycase-enhanced.sh
Nothing is passed
```

3. Write a script called mychmod using for utility to give execute permission to all files and directories in your home directory.

```
#!/bin/bash

files=$(ls $HOME)

for file in $files;
do
        echo $HOME/$file;
        chmod +x $HOME/$file
done
```

```
> ls -l $HOME
total 36
drwxr-xr-x  5 john john 16384 Feb 24 17:03 Desktop
drwxrwxr-x  2 john john  4096 Feb 24 17:02 Downloads
drwxrwxr-x 21 john john  4096 Feb 24 18:26 learning-notebook
drwxrwxr-x 12 john john  4096 Feb 22 16:43 pakinova
drw-rw-r--  4 john john  4096 Feb  8 23:13 Pictures
drwx------  4 john john  4096 Nov 12 20:00 snap
> ./mychmod.sh
/home/john/Desktop
/home/john/Downloads
/home/john/learning-notebook
/home/john/pakinova
/home/john/Pictures
/home/john/snap
> ls -l $HOME
total 36
drwxr-xr-x  5 john john 16384 Feb 24 17:03 Desktop
drwxrwxr-x  2 john john  4096 Feb 24 17:02 Downloads
drwxrwxr-x 21 john john  4096 Feb 24 18:26 learning-notebook
drwxrwxr-x 12 john john  4096 Feb 22 16:43 pakinova
drwxrwxr-x  4 john john  4096 Feb  8 23:13 Pictures
drwx--x--x  4 john john  4096 Nov 12 20:00 snap
>
```

4. Write a script called mybackup using for utility to create a backup of only files in your home directory.

```
#!/bin/bash

files=$(ls $HOME)

for file in $files;
do
        if [ -f $HOME/$file ];
        then
                echo "backing up file $HOME/$file"
                cp $HOME/$file $HOME/$file.bk
        fi
done
```

```
> rm $HOME/test
> touch $HOME/file{1..5}
> ls $HOME
Desktop  Downloads  file1  file2  file3  file4  file5  learning-
> chmod +x mybackup.sh
> ./mybackup.sh
> ./mybackup.sh
backing up file /home/john/file1
backing up file /home/john/file2
backing up file /home/john/file3
backing up file /home/john/file4
backing up file /home/john/file5
> ls
mybackup.sh  mycase-enhanced.sh  mycase.sh  mychmod.sh
> ls $HOME
Desktop    file1     file2     file3     file4     file5     lea
Downloads  file1.bk  file2.bk  file3.bk  file4.bk  file5.bk  pak
```

5. Write a script called mymail using for utility to send a mail to all users in the system.
Note: write the mail body in a file called mtemplate.

```bash
#!/bin/bash

template="mtemplate.txt"
usernames=$(awk -F: '$3 >= 1000 {print $1}' /etc/passwd )

for username in $usernames;
do
        echo "Sending mail to $username"
        mail $username < $template
done
```

```
> vi mymail.sh
> vi mtemplate.txt
> chmod +x mymail.sh
> ./mymail.sh
Sending mail to nobody
Sending mail to john
Sending mail to snapd-range-524288-root
Sending mail to snap_daemon
Sending mail to libvirt-qemu
Sending mail to someone
```

6. Write a script called chkmail to check for new mails every 10 seconds. Note: mails are saved in /var/mail/username.

```
> vi chkmail.sh
> chmod u+x chkmail.sh
> ./chkmail.sh
new mail
```

```bash
#!/bin/bash

old_lines_count=$(wc -l < /var/mail/$USER)

while true; do
  sleep 10
  new_lines_count=$(wc -l < /var/mail/$USER)

  if [ "$new_lines_count" -gt "$old_lines_count" ]; then
    echo "new mail"
    old_lines_count=$new_lines_count
  fi
done
```

7. What is the output of the following script

```
typeset –i n1
typeset –i n2
n1=1
n2=1
while test $n1 –eq $n2
do
        n2=$n2+1
        print $n1
if [ $n1 –gt $n2 ]
then
        break
else
        continue
fi
n1=$n1+1
print $n2
done
```

Check the loop condition, enter loop, print 1, exit loop when n2 > n1

8. Create the following menu:
    a. Press 1 to ls
    b. Press 2 to ls –a
    c. Press 3 to exit

Using select utility then while utility.

```
● john@pavilion:~/learning-notebook/Projects/08-bash/51-lab3$ ./menu.sh
  1) Press 1 to ls
  2) Press 2 to ls -a
  3) Press 3 to exit
  #? 0
  Invalid option
  #? 1
  menu.sh        myarr.sh  mybackup.sh         mycase.sh   mymail.sh
  mtemplate.txt  myavg.sh  mycase-enhanced.sh  mychmod.sh  mystery.sh
  #? 2
  .   menu.sh        myarr.sh  mybackup.sh         mycase.sh   mymail.sh
  ..  mtemplate.txt  myavg.sh  mycase-enhanced.sh  mychmod.sh  mystery.sh
  #? 3
```

```bash
#!/bin/bash

menu=("Press 1 to ls" "Press 2 to ls -a" "Press 3 to exit")

# select var in "${menu[@]}"
# do
# case $REPLY in
#   1) ls ;;
#   2) ls -a ;;
#   3) exit ;;
#   *) echo "Invalid option" ;;
# esac
# done

while true; do
for i in "${!menu[@]}"; do
  echo "$((i+1))) ${menu[i]}"
done

read -p "Enter choice: " choice

case $choice in
  1) ls ;;
  2) ls -a ;;
  3) exit ;;
  *) echo "Invalid option" ;;
esac
done
```

```
john@pavilion:~/learning-notebook/Projects/08-bash/51-lab3$ ./menu.sh
1) Press 1 to ls
2) Press 2 to ls -a
3) Press 3 to exit
Enter choice: 0
Invalid option
1) Press 1 to ls
2) Press 2 to ls -a
3) Press 3 to exit
Enter choice: 1
menu.sh        myarr.sh  mybackup.sh         mycase.sh   mymail.sh
mtemplate.txt  myavg.sh  mycase-enhanced.sh  mychmod.sh  mystery.sh
1) Press 1 to ls
2) Press 2 to ls -a
3) Press 3 to exit
Enter choice: 2
.   menu.sh        myarr.sh  mybackup.sh         mycase.sh   mymail.sh
..  mtemplate.txt  myavg.sh  mycase-enhanced.sh  mychmod.sh  mystery.sh
1) Press 1 to ls
2) Press 2 to ls -a
3) Press 3 to exit
Enter choice: 3
```

9. Write a script called myarr that asks a user how many elements he wants to enter in an array, fill the array and then print it.

```
#!/bin/bash

typeset -i n
typeset -a arr

read -p "Enter count of input: " n

for i in $(seq 0 $((n-1)));
do
        read -p "Enter element $((i+1)): " element
        arr[i]=$element
done

for e in ${arr[@]}
do
        echo $e
done
~
"myarr.sh" 17L, 212B
```

```
>./myarr.sh
Enter count of input: 5
Enter element 1: apple
Enter element 2: orange
Enter element 3: strawberry
Enter element 4: banana
Enter element 5: guava
apple
orange
strawberry
banana
guava
>
```

10.Write a script called myavg that calculates the average of all numbers entered by a user.
Note: use arrays

```
#!/bin/bash

read -p "Enter your numbers separated by spaces: " -a nums
sum=0
for num in ${nums[@]};
do
        sum=$((sum+num))
done

avg=$((sum/${#nums[@]}))
echo "Avg of your entered numbers is $avg"
```

```
>vi myavg.sh
>chmod +x myavg.sh
>./myavg.sh
Enter your numbers separated by spaces 1 2 3
Avg of your entered numbers is 2
>vi myavg.sh
>./myavg.sh
Enter your numbers separated by spaces: 5 10 15
Avg of your entered numbers is 10
>./myavg.sh
Enter your numbers separated by spaces: 100 50 0 25
Avg of your entered numbers is 43
```

11.Write a function called mysq that calculates square of its argument.

```
> vi mysq.sh
> chmod u+x mysq.sh
> ./mysq.sh 1 2 3
Square of 1 is 1
Square of 2 is 4
Square of 3 is 9
> ./mysq.sh
> ./mysq.sh 5
Square of 5 is 25
```

```
#!/bin/bash

for num in "$@"; do
        echo "Square of $num is $((num*num))"
done;
```